

Jordan Jordanov

Research Scientist & Software Developer — Computational Geometry, Applied ML, and Engineering Leadership

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14+ YEARS SOFTWARE DEVELOPMENT 6+ YEARS AI / MACHINE LEARNING 6+ YEARS ENGINEERING MANAGEMENT

■ PROFILE

Computer scientist and software developer with a PhD in computational geometry and strong experience in applied AI R&D. 14+ years across research, production engineering, and technical leadership. Contributor to CGAL, the largest open-source library for computational geometry. As CTO of an AI startup, scaled the engineering organization from 10 to 70, built an AIOps platform end-to-end, and delivered 50+ client-facing projects. Expert in C++ and Python. Fluent in five European languages.

■ EXPERIENCE

Researcher · **Braid Technologies** Tokyo, Japan Sep 2025 – Present

C++, Python, CGAL, VTK, libigl, Eigen, pybind11, CMake

- R&D on geometry generation and augmentation: mesh processing, quad remeshing, NURBS, and CAD interop for automated design.
- Contributed geometry algorithms and library-architecture improvements across the codebase.
- Shipped maintainable Python bindings (pybind11) for the core C++ geometry library, promoting faster prototyping across multiple teams.

Chief Scientist → CTO / Director · **Corpy & Co., Inc.** Tokyo, Japan Jul 2019 – Jul 2025

Python, PyTorch, NumPy, Docker, Celery, MySQL, JavaScript, Linux

- Architected and largely built a production AIOps platform from scratch (data management, annotation, augmentation, training, evaluation, deployment, monitoring), including edge deployment via model quantization and training/inference exposed through REST and gRPC for client integration.
- Ran a production computer-vision model zoo (object, pose, and depth detection, classification, segmentation) and built the MLOps from scratch: containerized CI/CD and a cloud-edge loop deploying quantized models to NVIDIA Jetson.
- Delivered 50+ client AI/ML projects end-to-end across manufacturing, automotive, healthcare, and industrial AI; owned the full lifecycle from scoping through staffing, QA, delivery, and support.
- Won 4 competitive government-funded research collaborations with a Japanese national research institute on AI quality-assurance guidelines (autonomous-driving vision, anomaly detection, fairness in financial risk, LLM QA); built a model-agnostic, task-extensible LLM evaluation framework.
- Scaled the engineering organization from 10 to 70 people; designed and ran the company OKR system from scratch, owning the full hiring pipeline and performance management, and drove board-approved ISMS adoption and ISO process standardization toward IPO readiness.

R&D Engineer · **Gamestream** Ludres, France Jan 2019 – Jun 2019

C++, OpenGL

- Developed a low-latency game-streaming client for Samsung TVs (full-HD at 60fps) using a proprietary SDK; researched frame upscaling based on neural networks.

PhD Candidate · **LORIA / Université de Lorraine** Nancy, France Jan 2016 – Dec 2018

C++, CGAL

- Formulated and implemented an algorithm for periodic Delaunay triangulations of the Bolza surface (the most symmetric compact hyperbolic surface of genus 2), now part of CGAL; published at SoCG 2017 and Curves & Surfaces 2018.

Graduate Research Assistant · **IACM-FORTH** Heraklion, Greece Dec 2014 – Oct 2015

Python, C++, VMTK

- Applied geometric processing to medical research on abdominal aortic aneurysms; developed a geometry-based method for quantifying morphological change, published in Medical & Biological Engineering & Computing.

Software Developer · **01Sistemi** Italy May 2013 – Jun 2014

Java, InstantDeveloper

- Built business software, including an Italian cadastral PDF parser and a tax-calculation engine for the 730 tax declaration.

Software Developer · **Nanosoft** Greece Dec 2009 – Mar 2012

VB.NET, WPF, Android

- Developed order-taking applications for Windows, Windows CE, and Android.

■ EDUCATION

- PhD, Computer Science** · **Université de Lorraine** · Nancy, France · 2016 – 2018
Thesis: Delaunay triangulations of a family of symmetric hyperbolic surfaces in practice
- MSc, Applied Mathematics** · **University of Crete** · Heraklion, Greece · 2014 – 2015
Thesis: Shape-Preserving Interpolation on the Sphere
- BSc, Applied Mathematics** · **University of Crete** · Heraklion, Greece · 2005 – 2013
Thesis: The Euclidean InSphere Predicate

■ SKILLS

Geometry & Math: Computational geometry · Delaunay triangulations · hyperbolic geometry · mesh processing · quad meshing / remeshing · NURBS · topology · CGAL · libigl · Eigen · VTK · PyVista · OpenCASCADE · FreeCAD

ML / AI: PyTorch · deep learning · computer vision · anomaly detection · object / pose / depth estimation · GANs · MLOps · model quantization · edge deployment · LLM evaluation · XAI · Quality assurance for AI (QA4AI)

Software Engineering: C++ · Python · pybind11 · CMake · Conan · UV · Docker · Celery · MySQL · Linux / Bash · full-stack web (JS) · Git · CI/CD · GCP / Azure / AWS (working familiarity)

Leadership: Org scaling (10→70) · OKR design and operation · full hiring pipeline · 50+ delivered projects · ISMS / ISO process standardization

Languages: Native: Greek, Bulgarian · Fluent: English, Italian · Advanced: French · Basic: Japanese

■ PUBLICATIONS

5. Cache-Property-Aware Features for DNS Tunneling Detection
N. Ishikura, D. Kondo, I. Iordanov, V. Vassiliades, H. Tode · ICIN, Paris, 2020
4. Systole of regular hyperbolic surfaces with an application to Delaunay triangulations
M. Ebbens, I. Iordanov, M. Teillaud, G. Vegter · Curves & Surfaces, Arcachon, 2018
3. Delaunay triangulations of regular hyperbolic surfaces
M. Ebbens, I. Iordanov, M. Teillaud, G. Vegter · Curves & Surfaces, Arcachon, 2018
2. Implementing Delaunay triangulations of the Bolza surface
I. Iordanov, M. Teillaud · SoCG, pp. 44:1–44:15, 2017
1. A novel approach for local abdominal aortic aneurysm growth quantification
E. Metaxa, I. Iordanov, E. Maravelakis, Y. Papaharilaou · Medical & Biological Engineering & Computing, 2016

■ OPEN SOURCE

CGAL — Computational Geometry Algorithms Library

- Contributed periodic Delaunay triangulations of the Bolza surface; announced February 2019 (cgal.org/2019/02/25/Hyperbolic_triangulations).